

Name _____ Surname _____ Date ____/____/____

Thermal physics · Mock midterm test · Version B

Time: 50 minutes · Total: 44 marks · Calculator, ruler and sharp pencil allowed. Section IV is in Italian: answer it in Italian or in English.

I · Multiple choice

[8]

Circle one letter for each question. Each question is worth 1 mark.

1. A student says: "This kettle contains a lot of heat." Which is the best correction?
 - A Nothing is wrong: a body stores heat in proportion to its temperature.
 - B The kettle contains heat only while it is hot, and none once it has cooled down.
 - C The kettle contains a lot of temperature, not heat.
 - D The kettle has a high temperature and a lot of internal energy; heat is energy being transferred, not energy stored.
2. A thermometer reads 25 °C when placed on a copper block, and again 25 °C when placed on a block of wood. The two blocks have never touched. What can be concluded?
 - A The two blocks are in thermal equilibrium with each other.
 - B The blocks have different temperatures, because copper conducts heat better.
 - C The blocks would exchange heat if they touched.
 - D Nothing, until the two blocks have touched.
3. Liquid nitrogen boils at 77 K. What is this temperature in degrees Celsius?
 - A 196 °C
 - B 350 °C
 - C -196 °C
 - D -77 °C
4. A fixed mass of gas at constant pressure has its volume plotted against temperature in °C. The straight line, extended backwards, reaches zero volume at about -273 °C. What does this suggest?
 - A The volume of any gas is proportional to its temperature in °C.
 - B There is a lowest possible temperature, 0 K.
 - C The experiment must be wrong, because a volume cannot be zero.
 - D All gases have the same density at -273 °C.
5. Group 1's thermistor thermometer changes by 6 counts for each °C, and Group 2's changes by 9 counts for each °C. Which statement is correct?
 - A Group 2's thermometer is more sensitive.
 - B Group 1's thermometer is more sensitive, because its number is smaller.
 - C Group 2's thermometer has a larger range.
 - D They are equally sensitive, because both used the same reference thermometer.
6. A copper wire 5.0 m long expands by 4.0 mm for a temperature rise of 50 °C. Another copper wire, 2.5 m long, is heated by 150 °C. By how much does it expand?
 - A 2.0 mm
 - B 12 mm
 - C 2.7 mm
 - D 6.0 mm
7. A steel ring is heated uniformly. Which statement about its inner diameter is correct?
 - A It decreases, because the ring gets thicker.
 - B It increases by the same fraction as the outer diameter.
 - C It stays the same, since only the metal expands.
 - D It increases by less than the outer diameter.
8. A solid sphere is heated, and its radius increases by 0.050 %. Approximately by what percentage does its volume increase?
 - A 0.050 %
 - B 0.10 %
 - C 0.15 %
 - D 0.0025 %

II · Alternative to practical

[14]

A group of students builds a thermometer from a thermistor and an Arduino board. The board gives a reading **N**, a number from 0 to 1023 with no unit (counts). The students calibrate the thermometer in four water baths. In each bath they measure the temperature **T** with a reference liquid-in-glass thermometer with 1 °C divisions, and they take five readings of **N**.

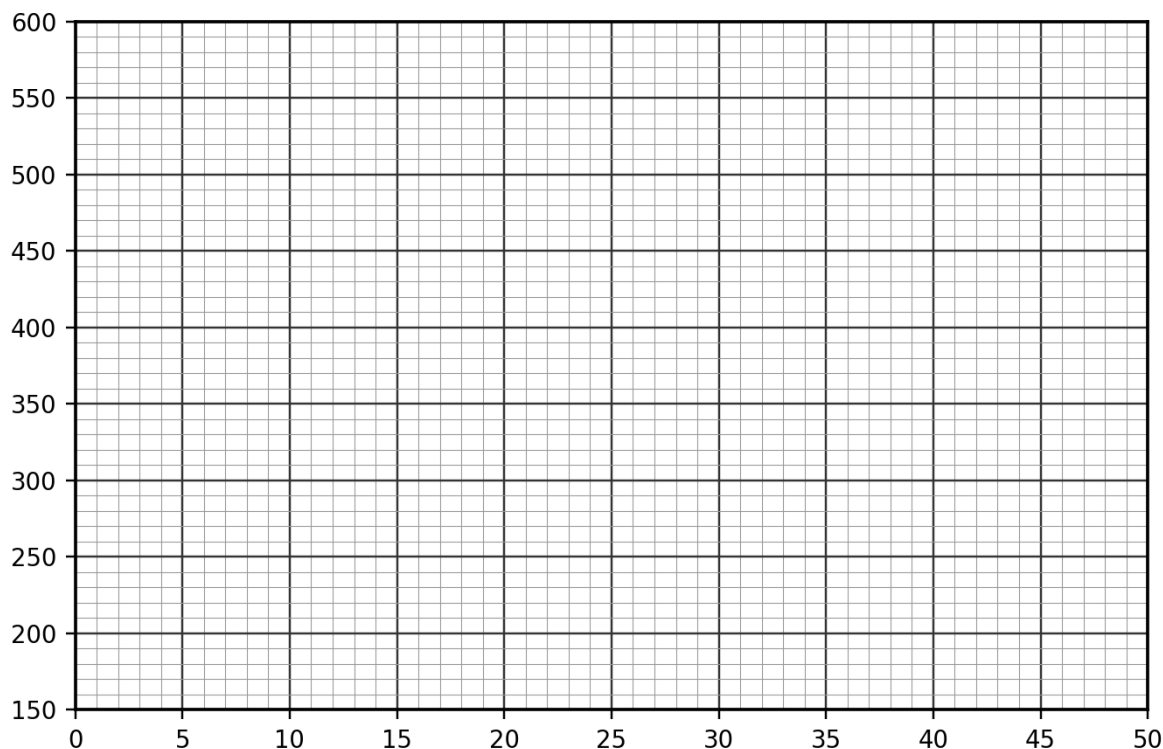
Bath	T / °C	N ₁	N ₂	N ₃	N ₄	N ₅	mean N
1 (ice water)	0.5	193	195	197	196	193	194.8
2 (cold water)	11.0	299	299	298	299	297	298.4
3 (room temperature)	22.5	390	426	392	391	393	
4 (hot water)	39.0	512	513	512	513	515	513.0

(a) State the independent variable and the dependent variable of this experiment. [2]

(b) One of the twenty readings in the table is anomalous. Circle it in the table and give one reason for your choice. [1]

(c) The mean N for one bath has been left blank. Calculate it, ignoring the anomalous reading, and write it in the table. [1]

(d) On the grid, plot the four mean values of N against T. The scales are printed. Mark each point with a small cross. Label both axes with the quantity and the unit. [3]



(e) Draw a single straight line of best fit with a ruler. It should pass as close as possible to all the points, with about as many points above it as below it. Do not join the points. [1]

(f) Use your line to find the gradient m (in counts per $^{\circ}\text{C}$). Show the triangle you use on the graph. [2]

(g) Read from your graph the value N_0 where the line crosses $T = 0^{\circ}\text{C}$. Then write the calibration equation in the form $T = (N - N_0) / m$ with your values. [1]

(h) The students now place the thermometer in a fifth bath, independent of the first four, and read $N = 429$. Use your equation to calculate the temperature of this bath. [1]

(i) The reference thermometer reads 30.6°C in the fifth bath. The client requires the Arduino thermometer to agree with the reference within $\pm 1^{\circ}\text{C}$. Does it meet the requirement? Show your calculation. [1]

(j) Suggest one specific improvement to the students' method that would make the calibration more reliable. [1]

III · Structured questions [14]

1. Temperature scales and heat

(a) Convert (i) 56°C into kelvin and (ii) 4 K into degrees Celsius. [2]

(b) A student says: "This radiator holds a lot of heat." Explain what is wrong with this statement. Use the words *temperature*, *heat* and *energy*. [2]

(c) Explain why a temperature of -5 K cannot exist. [1]

2. The thermometer

(a) State the two fixed points used to calibrate a Celsius thermometer, with their values. [2]

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- (b) A liquid-in-glass thermometer that has been in a hot room is plunged into iced water. In the first second the liquid column rises slightly before it starts to fall. Explain why. [2]

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3. Thermal expansion

Linear expansion coefficients, α ($\times 10^{-6} \text{ }^\circ\text{C}^{-1}$):

Material	steel	copper	brass	aluminium	glass
α	12	17	19	23	9

- (a) A copper pipe is 5.0 m long at $-5 \text{ }^\circ\text{C}$. Calculate its increase in length when it is heated to $35 \text{ }^\circ\text{C}$. [3]

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- (b) A plate of the same copper has an area of 0.60 m^2 at $-5 \text{ }^\circ\text{C}$. Calculate its increase in area when it is heated to $35 \text{ }^\circ\text{C}$. Use $\beta = 2\alpha$ for the area expansion coefficient. [2]
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Boccola di alluminio in un compressore. Una ditta che produce compressori per frigoriferi deve montare una boccola di alluminio (massa 0,35 kg, lunghezza 30 mm) su un albero d'acciaio. A 20 °C il foro della boccola ha diametro 40,00 mm e l'albero ha diametro 40,05 mm. La boccola viene scaldata in un forno a resistenza da 2,0 kW per circa 5 minuti, mentre l'albero resta a 20 °C; per infilarla senza forzare serve un gioco di almeno 0,04 mm, cioè il foro deve superare di 0,04 mm il diametro dell'albero. La lega di alluminio non va portata oltre 200 °C, altrimenti perde le sue proprietà. Il reparto monta 80 pezzi per turno e il compressore gira a 2900 giri/min.

Dati: coefficiente di dilatazione lineare dell'alluminio $23 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$; dell'acciaio $12 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$.

Chiamiamo *interferenza* la differenza (diametro dell'albero – diametro del foro) quando i due pezzi sono alla stessa temperatura: se è positiva, l'albero è più largo del foro e il collegamento è serrato.

(a) Scaldando la boccola, il diametro del foro aumenta, diminuisce o resta uguale? Spiega in una riga. [1]

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(b) A quale temperatura minima va portata la boccola perché scivoli sull'albero con il gioco richiesto? Verifica che il valore sia compatibile con il limite indicato dal costruttore. [3]

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(c) Di quanto aumenta, in percentuale, l'area della sezione del foro alla temperatura trovata in (b)? [2]

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(d) In esercizio la boccola e l'albero arrivano insieme a 150 °C. Calcola l'interferenza a quella temperatura e dì se il collegamento resta serrato, spiegando perché. [2]

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Fine della prova.